

# SAMPLE FINAL EXAMINATION

Sample

| Question               | Points | Score |
|------------------------|--------|-------|
| Basic concepts         | 56     |       |
| SVM and Kernel Methods | 9      |       |
| Neural Network         | 10     |       |
| PCA                    | 10     |       |
| Overfitting            | 15     |       |
| Total:                 | 100    |       |

- Please write down your **name** and **student ID** on the **answer paper**.
- The exam time is 2 hours.
- Even if you are not able to answer all parts of a question, write down the part you know. You will get corresponding credits to that part.

## Question 1 [56 points]: Basic concepts

(single choice just has one solution, while multiple choice has at least two solutions. In these choice questions and true or false questions, you *do not* need to justify your answer.)

- (a) [2 points] The least squares learning problem has a unique solution (when data matrix  $\mathbf{X} \in \mathbb{R}^{n \times d}$  has full column rank) or has no solution (when  $n < d$ ). (true or false)
- (b) [3 points] Is perceptron learning algorithm guaranteed to stop in a certain number of iterations? Briefly justify your answer.
- (c) [3 points] Describe the key difference between classification, regression, and clustering.
- (d) [2 points] Learning is possible in general. (true or false)
- (e) [2 points] Robust linear regression is more robust to outliers than least square. (true or false)
- (f) [3 points] Choose the true statement. (single choice)
  - (1) Linear model is linear in data sample  $\mathbf{x}$ .
  - (2) Linear model means that the learning problem is a linear optimization problem.
  - (3) None of them is true.
- (g) [3 points] Least squares is a maximum likelihood estimator. (true or false)
- (h) [3 points] To learn the underlying target function  $g$ , we can fit perfectly the training data  $\{(\mathbf{x}_i, y_i)\}$ . Then, the target  $g$  is learned. (true or false)
- (i) [3 points] Consider a hypothesis space  $\mathcal{H}$ . Suppose the growth function  $\mathcal{G}_{\mathcal{H}}(2) = 4, \mathcal{G}_{\mathcal{H}}(3) = 8, \mathcal{G}_{\mathcal{H}}(4) = 14, \mathcal{G}_{\mathcal{H}}(5) = 22, \mathcal{G}_{\mathcal{H}}(6) = 32$ . What is the VC dimension  $d_{VC}$  of  $\mathcal{H}$ ?
- (j) [3 points] From VC analysis,
  - i.* if we have more training data points (larger  $n$ ), then will the generalization error be larger or smaller?
  - ii.* if we choose a less complex  $\mathcal{H}$ , then will the generalization error be larger or smaller?
- (k) [3 points] Suppose the number of training samples  $n$  is fixed. Describe the process of making the out-of-sample error small from a VC analysis point of view.
- (l) [3 points] *i.* Write down the logistic regression model (binary case). *ii.* Suppose that you have learned a model  $\hat{\theta}$  for (binary) logistic regression. Now, we have a test data  $\mathbf{x}$ . Describe how can you classify  $\mathbf{x}$  based on the learned model.
- (m) [4 points] Choose the true statements about optimization algorithms. (multiple choice)
  - (1) GD utilizes first-order linear approximation of the learning problem.
  - (2) GD can be used to learn a soft-margin SVM classifier.
  - (3) Nesterov's accelerated GD (AGD) can be viewed as GD with a special step size.
  - (4) For large scale learning problem, SGD converges faster than GD.
  - (5) In one epoch, random reshuffling will use all the training data points for updating.
  - (6) For logistic regression learning problem, AGD converges slower than GD.
  - (7) SGD needs diminishing step sizes for convergence.
- (n) [3 points] List three components that may cause overfitting.
- (o) [3 points] Roughly describe the idea of validation.
- (p) [3 points] Choose the wrong statement about regularization. (single choice)
  - (1) Regularization is to estimate  $\text{Er}_{\text{out}}$ .
  - (2) Regularization helps reducing overfitting.
  - (3)  $\ell_2$ -regularized least square almost always admits a unique solution with properly chosen regularization parameter  $\lambda$ , even when the number of training data is less than the feature dimension.
  - (4) Proximal GD (PGD) has closed-form update when it is used to solve  $\ell_1$ -regularized logistic.

(q) [3 points] (true or false). AdamW is to apply Adam to the following regularized problem:

$$\min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta) + \frac{\lambda}{2} \|\theta\|_2^2, \quad \lambda > 0.$$

(r) [3 points] Choose the wrong statement.

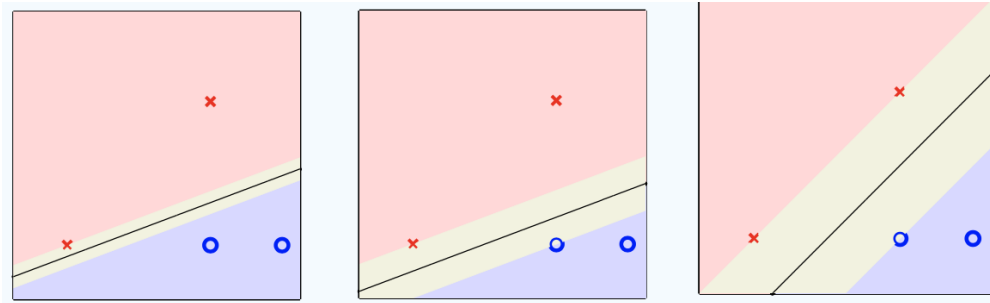
- (1) Neural network (NN) is a nonlinear model.
- (2) NN is to perform linear regression or linear classification in the final layer using the learned representation.
- (3) Backpropagation (BP) has a forward pass and backward pass.
- (4) In BP, we do not need to store anything in the forward pass.

(s) [4 points] Choose the wrong statements. (multiple choice)

- (1) Attention mechanism is used to capture semantic structure in a sentence.
- (2) Determining  $y_i$  of  $i$ -th token will only use input embedding  $x_i$  in attention.
- (3) Self-attention uses inner product between token's embedding to measure similarity.
- (4) Multi-head attention uses the same query, key, and value matrices among different heads.
- (5) Layer normalization does not have trainable parameters.
- (6) Language models usually use transformer to model conditional probability, and then predict the next token using autoregressive modeling.

## Question 2 [9 points]: SVM and Kernel Methods

(a) [2 points] Which one of the following classifiers is most likely learned by SVM? Briefly justify the reason.



(b) [3 points] Briefly describe why SVM classifier is robust for binary classification.

(c) [4 points] Derive the learning problem of kernelized soft-margin SVM with kernel function  $\kappa$ , by replacing  $\theta = \sum_{j=1}^n \alpha_j x_j$ , where  $\{(x_i, y_i)\}_{i=1}^n$  are the training data.

## Question 3 [10 points]: Neural Network

An  $L$  layer deep forward neural network is

$$f(x) = h(W_L \sigma(W_{L-1} \cdots \sigma(W_2 \sigma(W_1 x))),$$

where  $x \in \mathbb{R}^{d_0}$  is the input,  $W_l \in \mathbb{R}^{d_l \times d_{l-1}}$  is the weight of  $l$ -th layer,  $\sigma(\cdot)$  and  $h(\cdot)$  are the activation functions.

(a) [2 points] Suppose the input data  $x$  is of dimension 10 and the output label  $y$  is of dimension 1. Consider the 3 hidden layer neural network (with no bias terms) with neurons of  $\{12, 24, 8\}$ . Calculate the total parameter number for the neural network.

- (b) [4 points] Consider the scalar input  $x \in \mathbb{R}$ . Let  $h(\cdot)$  be the identity function and  $\sigma(\cdot)$  be the Relu activation function:

$$h(x) = x, \quad \sigma(x) = \max\{x, 0\}.$$

Design a neural network  $f(x) = x$ . Specify the weight of each layer.

(Hint:  $x = \max\{x, 0\} - \max\{-x, 0\}$ )

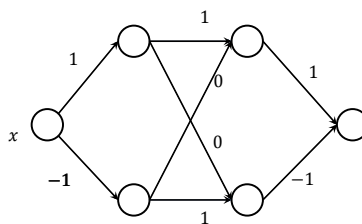


Figure 1: Neural network design for problem (b).

- (c) [4 points] Consider the input  $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$ . Let  $h(\cdot)$  be the identity function and  $\sigma(\cdot)$  be the piece-wise ReLU activation function:

$$h(\mathbf{x}) = \mathbf{x}, \quad \sigma(\mathbf{x}) = \begin{bmatrix} \max\{x_1, 0\} \\ \max\{x_2, 0\} \\ \vdots \\ \max\{x_{d_0}, 0\} \end{bmatrix}.$$

Design a neural network  $f(\mathbf{x}) = \max\{x_1, x_2\}$ . Specify the weight of each layer.

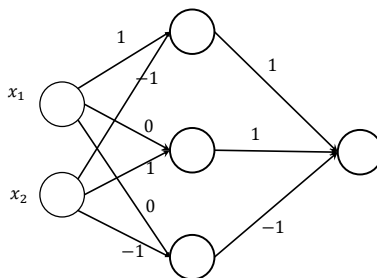


Figure 2: Neural network design for problem (c).

#### Question 4 [10 points]: PCA

Consider the following form of PCA:

$$\underset{\mathbf{A}^\top \mathbf{A} = \mathbf{I}}{\text{maximize}} \sum_{i=1}^n \|\mathbf{A}^\top \mathbf{x}_i\|_2^2,$$

where  $\mathbf{x}_i \in \mathbb{R}^d$  and  $\mathbf{A} \in \mathbb{R}^{d \times k}$ .

- [3 points] How to compute the solution  $\mathbf{A}$ ?
- [3 points] After computing  $\mathbf{A}$ , how to compute the principle component matrix  $\mathbf{\Theta}$ ? What is the dimension of  $\mathbf{\Theta}$ ?
- [4 points] Suppose we increase  $k$ , will the error  $\sum_{i=1}^n \|\mathbf{x}_i - \mathbf{A}\boldsymbol{\theta}_i\|^2$  be smaller or larger? Here,  $\mathbf{A}$  is the solution in Part (a) and  $\boldsymbol{\theta}_i$  is the  $i$ -th column of  $\mathbf{\Theta}$  in Part (b). Briefly justify the answer.

### Question 5 [15 points]: Overfitting

We will discuss overfitting, validation, and regularization (weight decay) in this question.

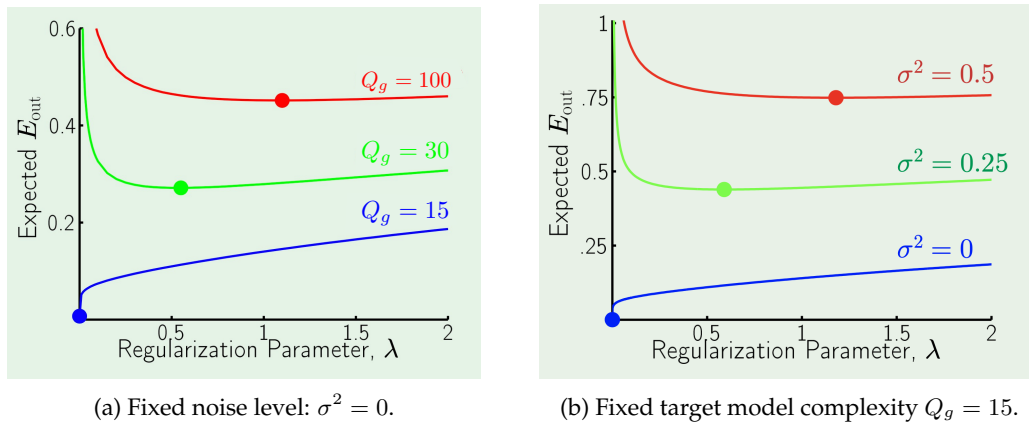


Figure 3: Evolution of  $E_{\text{out}}$  with respect to the regularization parameter  $\lambda$ , where the optimal  $\lambda$  with lowest  $E_{\text{out}}$  in each curve is dotted. In the figure,  $Q_g$  is the target model complexity (i.e., the underlying target model  $g$ 's complexity), and  $\sigma^2$  is the noise level in the training data. In both Figure 1 (a) and Figure 1 (b), we use a model  $f$  with complexity  $Q_f = 15$  to fit the training data using weight decay regularization with a regularization parameter  $\lambda \geq 0$ .

- [6 points] Consider Figure 3 (a).
  - Explain the reason why the blue curve ( $Q_g = 15$ ) increases  $E_{\text{out}}$  with increasing  $\lambda$ .
  - Explain the reason why the optimal  $\lambda$  increases with increasing target complexity  $Q_g$ .
- [5 points] Consider Figure 3 (b).
  - In the blue curve, is the model underfit, goodfit, or overfit when  $\lambda = 1$ ?
  - Explain the reason why the green (or red) curve first decrease to the lowest  $E_{\text{out}}$ , then increase  $E_{\text{out}}$ .
- [4 points] Suppose that we are in the regime of binary linear classification and the definitions of  $E_{\text{in}}$  and  $E_{\text{out}}$  are both using binary error. In the context of validation, show the following inequality appeared in our lecture slides, where  $k$  is the validation dataset size:

$$E_{\text{out}}(f') \leq E_{\text{val}}(f') + \mathcal{O}\left(\frac{\log(1/\delta)}{\sqrt{k}}\right),$$

which holds true with probability at least  $1 - \delta$ . Here,  $f'$  is trained on the  $n - k$  new training set.